

R1: SNR Axis and Measured Noise

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Over what signal-to-noise range is the inverse Identifiable, and where do the real sensors sit? Amir Dollie (DLLAMI004),
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SHARC buoy project Code: `runR1SNR.m`, `r1Boundary.m`, `plotR1SNR.m` Results: `Results/R1/R1_20261008-185607` (commit 0389536)

Rerun after the amplitude amendment (8 Oct). The EMM FRFs used by the 3 Oct run were low by $\alpha(\omega) = H\omega^2/g$; `computeSensorFRF` now multiplies by α and the twin cache was migrated exactly (`cfg.meta.changes`, 2026-10-08). Settings unchanged. This summary supersedes R1_20261003_135701. The signal is now correctly scaled, so every noise level corresponds to a higher SNR than before; the window has moved towards noisier sensors and its noisy edge has left the swept range. (`plotR1SNR` was fixed for the new “Not identifiable” band: colour lookup by class name.)

Summary

Question. With the floe (p_0), layout (Level 2C, reference s26), sea (JONSWAP, $H_s = 0.05$ m, $\omega_p = 5$ rad/s, $\gamma_J = 3.3$), record ($N = 65536$) and estimator ($L = 2048$) fixed, sweep the accelerometer noise floor S_{nn} over four decades and locate every class boundary. **Outcome.** On the whole sweep the inverse is **bias-limited, never precision-limited**: $\sigma_{\ln \beta} \leq 0.41\%$ even at $2555 \mu g/\sqrt{\text{Hz}}$. It is Identifiable for

$$S_{nn} \geq 4.536 \times 10^{-7} (\text{m/s}^2)^2 \text{ s/rad} \iff \text{ASD} \geq 172.1 \mu g/\sqrt{\text{Hz}}$$

up to the end of the sweep; Marginal between 77.6 and $172.1 \mu g/\sqrt{\text{Hz}}$ (d_W from 1 to 0.5); and Not identifiable below $77.6 \mu g/\sqrt{\text{Hz}}$ ($d_W > 1$). Five of the six noise cases are Identifiable, including the **LSM6DSV16X** ($d_W = 0.436$) and the **ICM-42688-P**; the datasheet-grade case is Not identifiable through bias ($d_W = 1.21$) despite the best precision ($\sigma_{\ln \beta} = 0.020\%$). For the LSM6DSV16X at this spectral shape, the sea must not exceed $H_s = 0.066$ m at $L = 2048$.

1 What the study does

One axis. The system is linear and the signal spectrum scales as H_s^2 , so for a fixed spectral shape only the ratio H_s^2/S_{nn} matters; one S_{nn} sweep is equivalently an H_s sweep (tester: scaling S_{nn} by 4 and H_s by 2 changes σ , d_W and F by less than 10^{-9}). Everything except S_{nn} is the E1 default; the feature f_0 , the Jacobian J and the Welch feature bias b_W are identical along the axis, only $F = \sum_k J_k^T \Sigma_k^{-1} J_k$ changes.

Why the quiet end fails. Lowering the noise raises F and so shrinks σ . The Welch feature bias b_W does not depend on the noise, but its significance $d_W = \sqrt{\delta \theta_W^T F \delta \theta_W}$ is measured against the shrinking uncertainty. A quiet enough sensor therefore turns a fixed estimator bias into a significant one.

Method (pre-registered in `cfg.R1`, clarification logged before the first run). (1) 41 values of S_{nn} from 10^{-8} to 10^{-4} per rad/s, 0.1 decade apart (25.6 to $2555 \mu g/\sqrt{\text{Hz}}$). (2) The six named noise cases evaluated exactly. (3) Every class change between neighbouring grid points refined by bisection in $\log_{10} S_{nn}$ on the metric that crosses its threshold, to 10^{-3} decade (9 evaluations each). (4) Each boundary also expressed as the H_s at which the LSM6DSV16X would have that SNR, $H_{s,\text{eq}} = 0.05 \sqrt{S_{\text{LSM}}/S_{nn}}$ m. ASD in $\mu g/\sqrt{\text{Hz}}$ is $\sqrt{2\pi S_{nn}}/9.81 \times 10^{-6}$. Runtime 87 s, no EMM solves.

2 Results

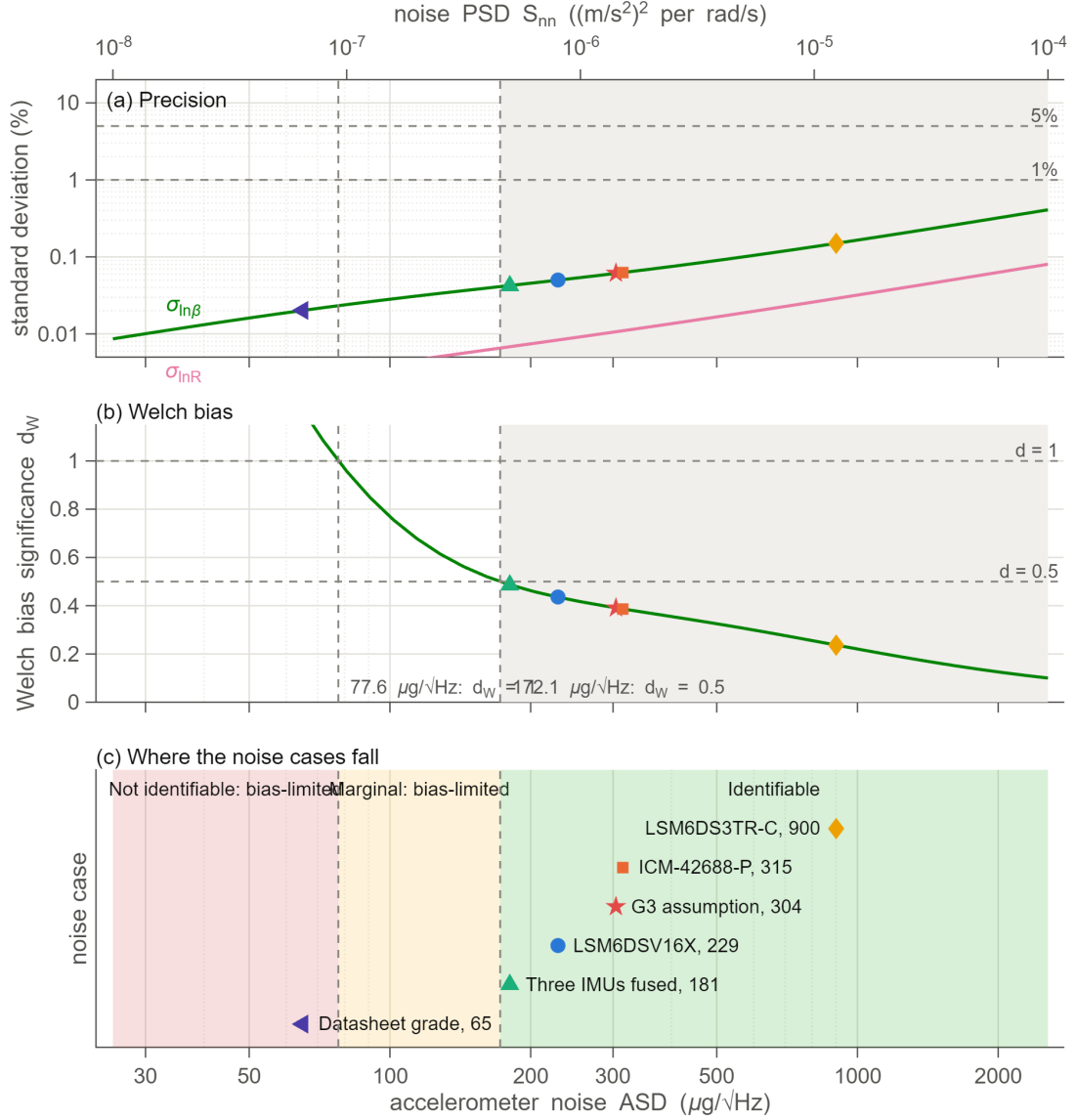


Figure 1. The SNR axis for p_0 , Level 2C, $N = 65536$, $L = 2048$ (corrected FRF). Bottom axis: noise ASD; top axis: the same noise as a PSD per rad/s. (a) $\sigma_{\ln\beta}$ and $\sigma_{\ln R}$ with the 1% and 5% limits. (b) Welch bias significance with the $d = 0.5$ and 1 limits. (c) The six noise cases over the class band of the sweep. Dashed verticals: the refined boundaries at 77.6 and 172.1 $\mu\text{g}/\sqrt{\text{Hz}}$; shaded in (a) and (b): the Identifiable range.

2.1 Three regimes, all set by the bias

Regime	ASD ($\mu\text{g}/\sqrt{\text{Hz}}$)	S_{nn} (per rad/s)	Class	Binding metric
Strongly bias-limited	25.6 to 77.6	10^{-8} to 9.214×10^{-8}	Not identifiable	d_W 3.28 down to 1
Bias-limited	77.6 to 172.1	9.214×10^{-8} to 4.536×10^{-7}	Marginal	d_W 1 down to 0.5
Identifiable	172.1 to >2555	4.536×10^{-7} to $> 10^{-4}$	Identifiable	none

$\sigma_{\ln\beta}$ runs from 0.009% to 0.409% over the sweep and never reaches 1%. Extrapolating $\sigma_{\ln\beta} \propto \sqrt{S_{nn}}$ from the noisy end puts the precision limit near $S_{nn} \approx 6 \times 10^{-4}$ (about 6000 $\mu\text{g}/\sqrt{\text{Hz}}$), outside the registered range and not claimed. $\kappa(F)$ stays between 67 and 154; no bin is rejected anywhere.

2.2 Where the noise cases sit

Noise case	ASD ($\mu\text{g}/\sqrt{\text{Hz}}$)	PSD/G3	ASD/G3	$\sigma_{\ln\beta}$	$\sigma_{\ln R}$	d_W	Class	max H_s (m)
Datasheet grade	65.1	0.046	0.214	0.020%	0.003%	1.209	Not identifiable (bias)	0.019
Three IMUs fused	180.7	0.353	0.594	0.043%	0.007%	0.487	Identifiable	0.053
LSM6DSV16X	228.5	0.565	0.752	0.050%	0.008%	0.436	Identifiable	0.066
G3 assumption	304.1	1.000	1.000	0.061%	0.011%	0.391	Identifiable	0.088
ICM-42688-P	315.0	1.073	1.036	0.063%	0.011%	0.387	Identifiable	0.092
LSM6DS3TR-C	899.8	8.757	2.959	0.150%	0.029%	0.237	Identifiable	0.261

max H_s : the largest significant wave height of this spectral shape at which the sensor stays Identifiable at $L = 2048$, $0.05\sqrt{S/S_{10}}$ m with $S_{10} = 4.536 \times 10^{-7}$. The smallest usable H_s lies below the sweep (for the LSM6DSV16X, below 0.0045 m). Pre-amendment (3 Oct), LSM6DSV16X: $\sigma_{\ln\beta} = 0.355\%$, $d_W = 0.309$, window 110 to 654 $\mu\text{g}/\sqrt{\text{Hz}}$.

3 Interpretation

1. **Bias, not noise, sets the class.** With the signal correctly scaled, every realistic sensor gives $\sigma_{\ln \beta}$ well under 1%; the only failures are on the quiet side, through the $L = 2048$ Welch bias. The two-sided SNR window of the first run is now one-sided within the registered range.
2. **The LSM6DSV16X has a small bias margin.** It sits a factor 1.33 in ASD above the bias boundary ($d_W = 0.436$). The fused three-IMU estimate sits a factor 1.05 above it ($d_W = 0.487$): fusion buys 14% in $\sigma_{\ln \beta}$ and spends almost all of the remaining bias budget. At $L = 2048$ fusion is not worthwhile.
3. **In wave height: an upper limit only.** For the LSM6DSV16X the basin sea must stay below $H_s \approx 0.066$ m at this spectral shape; the reference 0.05 m is three quarters of the way there. Larger waves carry more information, but at $L = 2048$ the Welch bias becomes the limiting error. The remedy is the estimator, not the sea: a longer segment ($L = 4096$, D2) cuts the Welch bias several-fold.
4. **A quieter sensor is not worse in information.** The datasheet-grade case has the smallest uncertainty of all ($\sigma_{\ln \beta} = 0.020\%$); it is Not identifiable only because the fixed Welch bias is now more than one uncertainty unit. The quiet boundary is a property of the estimator setting, not a requirement on the hardware.
5. **Consistent with D2.** D2 found every sensor bias-limited along the record-length axis; R1 shows the same along the SNR axis. Both say the same thing: once the experiment is precise, the estimator sets the limit.
6. **The G3 assumption was realistic.** The ICM-42688-P sits almost exactly on it (1.07 in PSD) and the LSM6DSV16X is 25% quieter in ASD. The noisiest sensor, the LSM6DS3TR-C, is now Identifiable (0.150%); it was precision-limited before the amendment.

Limitations. One floe and one layout (p_0 , Level 2C); A1 showed the precision depends on the floe. One spectral shape (R2 covers shape; the H_s equivalence holds only at fixed shape). One record length and one L : the quiet boundary is specific to $L = 2048$ and $N = 65536$. White, known noise (R3 covers a misspecified floor). Analytic expectations, not realised records (M1 tests selected cases with Monte Carlo records). Model truncation bias (A1, V1) is outside this within-model analysis.

Requirement for the experiment and the thesis

For p_0 , Level 2C, $N = 65536$ and $L = 2048$, every sensor noisier than $172 \mu g/\sqrt{\text{Hz}}$ is Identifiable up to at least $2555 \mu g/\sqrt{\text{Hz}}$; quieter sensors, or larger waves, make the inverse bias-limited. The LSM6DSV16X ($229 \mu g/\sqrt{\text{Hz}}$) and the ICM-42688-P ($315 \mu g/\sqrt{\text{Hz}}$) are both Identifiable; for the LSM6DSV16X the sea should not exceed $H_s \approx 0.066$ m at the reference shape unless L is increased. Do not fuse IMUs at $L = 2048$. **M1 hand-off:** p_0 at LSM6DSV16X noise (Identifiable, $d_W = 0.436$).

Outputs (Results/R1/): R1_20261008.185607.log/.mat/_summary.txt; R1_sweep.csv, R1_sensor_points.csv, R1_boundaries.csv; figures/R1_fig1_snr_window (.pdf/.png). Superseded: R1_20261003_135701 (pre-amendment).